

Yi Chen

Ph.D. Candidate in Machine Learning
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Education

Ph.D. Electrical and Computer Engineering, University of Wisconsin-Madison, 2022-May 2028 (anticipated)
M.S. Electrical and Computer Engineering, University of Wisconsin-Madison, 2022-2025
B.A. Computer Sciences *Honors in Major*, University of Wisconsin-Madison, 2019-2022
B.A. Mathematics, University of Wisconsin-Madison, 2019-2022

Research Areas

Foundation Model Alignment: Developing methods for aligning LLMs with human values through preference learning and RLHF

Trustworthy AI: Mitigating hallucinations and ensuring factuality in generative models via conformal prediction

Crowdsourcing and Human-in-the-Loop ML: Efficient knowledge transfer from humans to machines through active learning and crowdsourcing

Working Experience

Summer 2026 Software Engineer Intern, Machine Learning - Core Ads Growth Eng Team, Meta

Preprints

2026 **Chen, Y.**, Chen, D., Chikodikar, S. M., Yin, C. H., and Vinayak, R. K. “Is Conformal Factuality for RAG-based LLMs Robust? Novel Metrics and Systematic Insights” *arXiv preprint*

Conference Publications

2025 Tatli, G., **Chen, Y.**, Mason, B., Nowak, R. D., and Vinayak, R. K. “Metric Learning in an RKHS.” *The 41st Conference on Uncertainty in Artificial Intelligence (UAI 2025)*

2025 Vishwakarma, H.* , **Chen, Y.***, Namburi, S.S.S., Tay, S.J., Vinayak, R. K., and Sala, F. “Rethinking Confidence and Thresholds in Pseudolabeling-based SSL.” *Forty-second International Conference on Machine Learning (ICML 2025)*

2025 **Chen, Y.** and Vinayak, R. K. “Query Design for Crowdsourced Clustering: Effect of Cognitive Overload and Contextual Bias.” *The Web Conference 2025 (TheWebConf 2025)* [Oral]

2025 Chen, D., **Chen, Y.**, Rege, A., Wang, Z., and Vinayak, R. K. “PAL: Pluralistic Alignment Framework for Learning from Heterogeneous Preferences.” *The Thirteenth International Conference on Learning Representations (ICLR 2025)*

- 2024 Vishwakarma, H., **Chen, Y.**, Tay, S. J., Namburi, S. S. S., Sala, F., and Vinayak, R. K. “Pearls from Pebbles: Improved Confidence Functions for Auto-labeling.” *The Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS 2024)*
- 2024 Tatli, G., **Chen, Y.**, and Vinayak, R. K. “Learning Populations of Preferences via Pairwise Comparison Queries.” *Proceedings of The 26th International Conference on Artificial Intelligence and Statistics (AISTATS 2024)*
- 2023 **Chen, Y.**, Vinayak, R. K., and Hassibi, B. “Crowdsourced Clustering via Active Querying: Practical Algorithm with Theoretical Guarantees.” *The Eleventh AAAI Conference on Human Computation and Crowdsourcing (HCOMP 2023)*

Workshop Publications

- 2025 **Chen, Y.**, Yin, C. H., Chikodikar, S. M., and Vinayak, R. K. “On the Scoring Functions for RAG-based Conformal Factuality.” *ICML 2025 Workshop on Reliable and Responsible Foundation Models (ICML 2025)*
- 2024 **Chen, Y.** and Vinayak, R. K. “Query Design for Crowdsourced Clustering: Effect of Cognitive Overload and Contextual Bias.” *Workshop on Models of Human Feedback for AI Alignment (ICML 2024)*
- 2023 Tatli, G., **Chen, Y.**, and Vinayak, R. K. “Learning Populations of Preferences via Pairwise Comparison Queries” *MFPL Workshop at the 40th International Conference on Machine Learning (ICML 2023)*
- 2023 **Chen, Y.**, Vinayak, R. K., and Hassibi, B. “Crowdsourced Clustering via Active Querying: Practical Algorithm with Theoretical Guarantees.” *AI & HCI Workshop at the 40th International Conference on Machine Learning (ICML 2023)*
- 2023 Tatli, G., **Chen, Y.**, and Vinayak, R. K. “Learning Preference Distributions From Pairwise Comparisons.” *9th International Workshop on Computational Social Choice (COMSOC 2023)*

Teaching Experience

- Fall 2025 ECE 539 - Introduction to Artificial Neural Networks
- Summer 2025 ECE 539 - Introduction to Artificial Neural Networks

Technical Skills

Languages: Python (primary), C/C++, Java, JavaScript, Rust, CUDA, SQL

ML/AI Frameworks: PyTorch (primary), TensorFlow, JAX, Hugging Face Transformers, scikit-learn

Tools & Platforms: AWS (Lambda, S3, EC2), Docker, Git, MongoDB, React, Node.js

Specializations: Machine Learning, Deep Learning, Reinforcement Learning (RLHF), Crowdsourcing

Professional Service & Awards

Reviewer: COLM 2026, UAI 2026, ICLR 2026, ICLR 2025, TheWebConf 2025, NeurIPS 2024 (workshops), ICML 2024 (workshops)

Awards: HCOMP/CI 2023 Travel Award, UW-Madison Conference Presentation Award

Industry Collaboration Potential

Experience with distributed systems and production ML pipelines

Theoretical background combined with practical implementation skills

Research directly applicable to LLM alignment, RLHF, and trustworthy AI systems

Proven ability to collaborate across teams (6 different co-author groups)

Updated May 2026